Last 2 Sections on the document deliverable:

Kylie Lynch

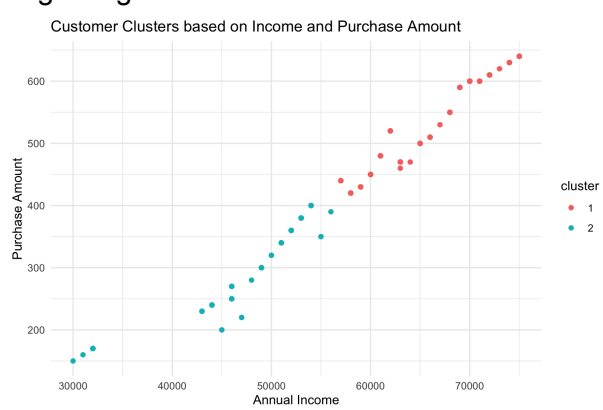
Progress So Far

● Completed Work: Describe what you have accomplished so far. Have you created any models, or completed any analyses?

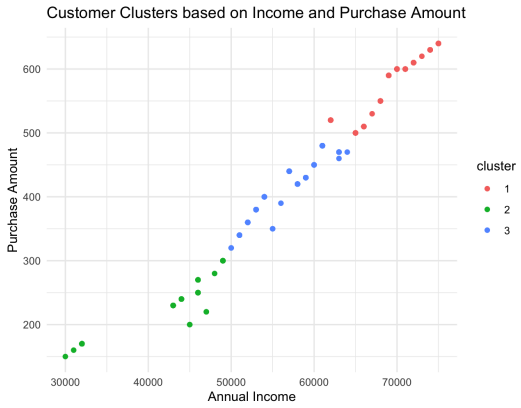
So far, we have made significant progress in preparing and analyzing our dataset, as well as generating a few completed models. We started by cleaning the data, ensuring consistency by transforming categorical variables like geography into factors, addressing missing values, and looking for duplicates. No missing values or duplicates were found, so it was not necessary to omit any data. In addition to this, we standardized column forms by converting region names to lowercase to provide consistency across the analysis, given that there were some uppercase regions and those same regions in lowercase. Following the data processing, we segmented by age group to investigate how age affects purchasing patterns, and we calculated summary statistics for each group, such as average income, purchase amount, and loyalty score.

We were also able to implement a K-means clustering model to discover data-driven customer segments based on income, purchase amount, and loyalty scores. The data was already well formatted, so there was no need to change algorithms or clustering models at this point in the project. To improve the clustering results, we scaled the numeric variables to ensure that all features contributed equally. We used a three-cluster model to find significant patterns after experimenting with a two-cluster model. The additional cluster we added helped reduce the potential for missing important patterns or factors. With only two clusters, different types of customers, such as moderate spenders, might have been grouped incorrectly, masking their distinct behavior.

Beginning 2 cluster model:



After changing to a 3-cluster model:

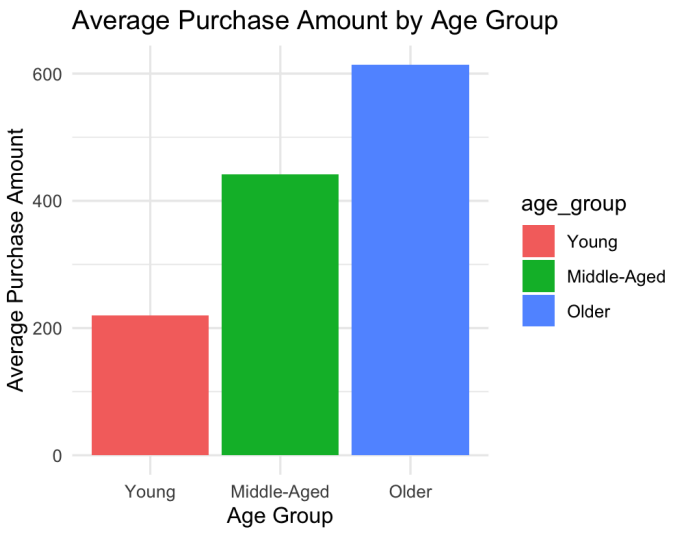


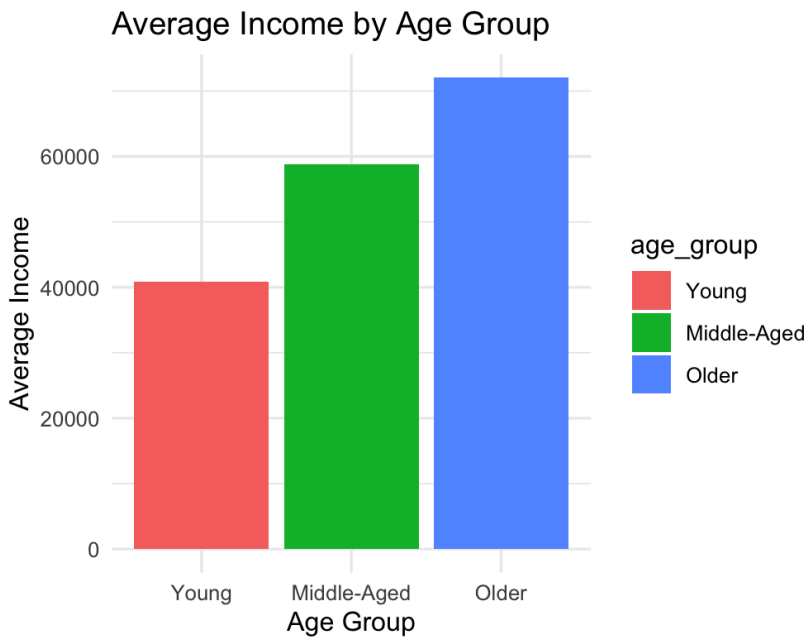
● Preliminary Results: Share any early findings, results, or insights you have uncovered.

Our early analyses have revealed several key insights. The age group segmentation showed that young customers (0–30 years) tend to have lower income, spend less, and exhibit lower loyalty compared to older age groups. This is consistent with life stages, as individuals tend to earn more as they gain experience and seniority in the workforce. Middle-aged (31–50) and Older (50+) customers demonstrated higher purchasing power, with the older customers also showing the highest loyalty scores. Key conclusions we have found from our analysis so far include that income and spending typically increase with age as well as loyalty to a brand, and the largest age group consists of middle-aged customers, which is helpful to know for marketing purposes. They can be targeted more during marketing campaigns, which may produce more sales for ABC Sales Inc. These findings suggest that age influences both spending behavior and brand engagement.

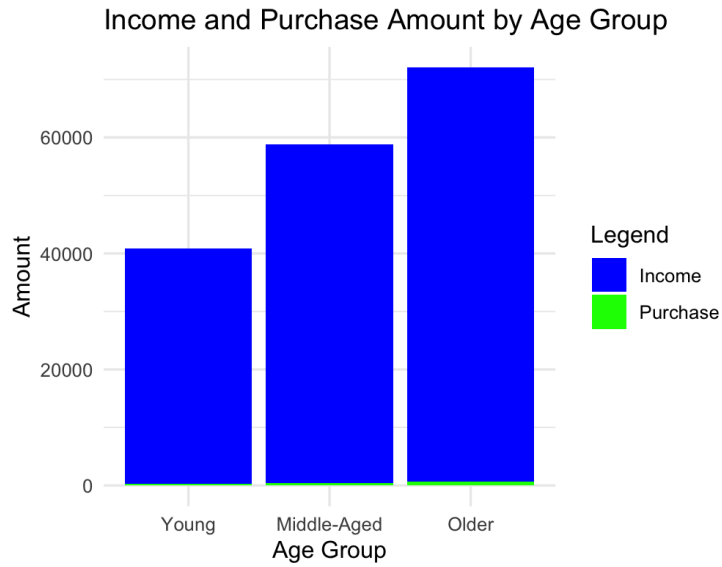
The K-means clustering analysis yielded three distinct clusters. The first cluster consists of high-income, high-spending customers, mostly older individuals. The second cluster comprises low-income, low-spending customers, primarily young individuals, indicating a need for targeted discounts and loyalty incentives. The third cluster represents mid-income, moderate-spending customers, mainly middle-aged customers, suggesting potential for cross-selling or personalized offers. We also decided to cross-tabulate age groups with the clusters and found that all young customers (0–30 years) belong to Cluster 2 (low income, low purchases), indicating they are price-sensitive and may benefit from discounts or student-focused offers. The middle-aged group (31–50 years) is the most diverse, with customers spread across all three clusters: 39 in Cluster 1 (high income, high purchases), 8 in Cluster 2 (low income, low purchases), and 96 in Cluster 3 (mid-income, moderate purchases). This group may benefit best from personalized offers. Lastly, all older customers (50+ years) belong to Cluster 1, making them the most valuable segment to our business sales, and they should be targeted with exclusive offers, luxury products, or loyalty programs. These insights have already started to shape our marketing strategy recommendations.

● Visualizations: Include any relevant visualizations (graphs, charts, tables) that show your preliminary results or the current state of your project.

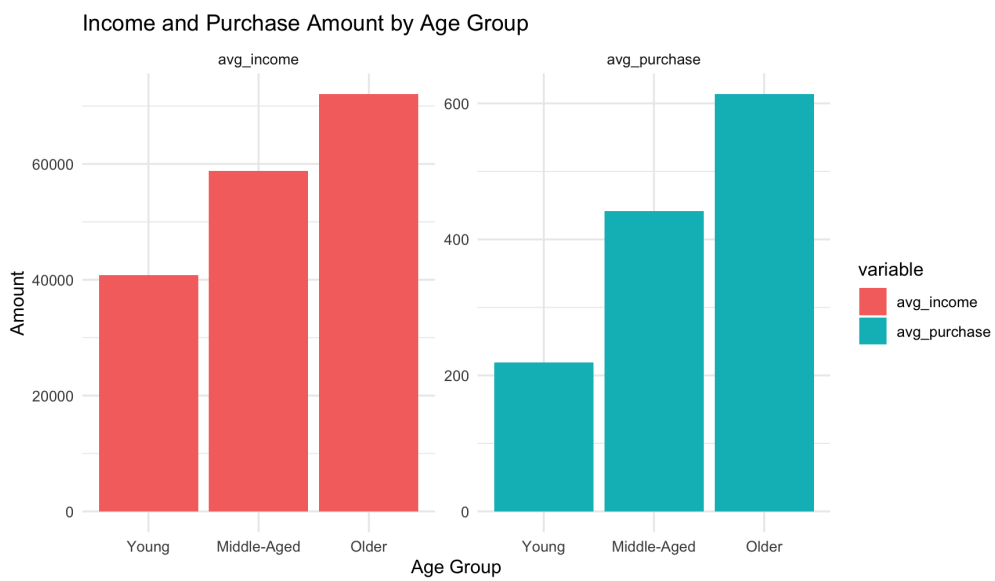

Average Purchase Amount by Age Group
Average Purchase Amount
600
400
200
0
Young
Middle-Aged
Older
Age Group
age_group
Young
Middle-Aged
Older


Average Income by Age Group
Average Income
60000
40000
20000
0
Young
Middle-Aged
Older
Age Group
age_group
Young
Middle-Aged
Older

Income vs Purchases (not most important graph; does not provide enough data to analyze)



Combined graphs:



Screenshot of cross-table: age groups with the clusters

```
> table(customerpurchasing$age_group, customerpurchasing$cluster)

              1  2  3
  Young       0 51  0
  Middle-Aged 39  8 96
  Older       44  0  0
>
```

—---------- Can also add the clustering graphs here—----------

Timeline & Next Steps:

● Future Plan: What remains to be done? Provide a brief timeline of your remaining tasks.

In the next phase, we will focus on analyzing the regional impact to understand how geography influences purchasing patterns and customer segments. This analysis will help uncover potential variations in customer behavior across different regions, providing deeper insights into market-specific strategies. Following the regional analysis, we will combine our findings from age, income, region, and cluster analysis, in order to draw more meaningful conclusions about customer behaviors and develop targeted marketing strategies. Our overall goal is to provide actionable recommendations that align with the needs of both our customers and stakeholders. This synthesis will be completed by Week 12 of the semester. In Weeks 12–13, we will finalize all visualizations to clearly communicate the key insights in our final report. During the same period, we will also compile the full report, making sure that all findings, conclusions, and recommendations are well-structured and aligned with our research question. Lastly, we will design the presentation in Week 13 to deliver a polished and engaging summary of our work.

● Challenges: Highlight any challenges or obstacles your group may face, and how you plan to address them.

One potential problem we may face is managing the complexity of the regional study, which could include a large number of interacting factors. To overcome this, we will utilize pivot tables, visualizations, and clustering approaches to locate and simplify patterns in our data, as we have done previously. Another concern is translating the clustering results into a meaningful marketing strategy. We will ensure that our recommendations are particular and relevant by concentrating on how client segments relate to the demands of our stakeholders. We also predict that time management and group coordination will be critical, particularly since some tasks are dependent on the completion of others. To stay on track, we will schedule weekly check-ins to address any roadblocks before the final presentation. With this approach in place, we are confident as a team in staying on track and analyzing data that will help us enhance our marketing strategy for ABC Sales Inc.